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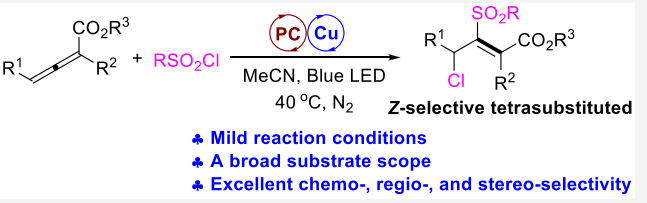
Photo- and Copper Dual-Catalyzed *Z*-Selective Chlorosulfonylation of Allenes for Tetrasubstituted Alkenes

Yaqi Shi, Shichao Hong, Xue Zhang,* Jian Zheng,* and Shengming Ma*

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ABSTRACT: The selectivity control of C═C double bonds, especially in tetrasubstituted olefins, is a fundamental issue due to their importance in organic chemistry, medicinal chemistry, and material chemistry. Herein, we observed that the stereoselectivity of chlorosulfonylation of allenoic acids or allenoates was reversed by applying an organic photocatalyst in combination with a copper catalyst $Cu(acac)_2$, affording *Z*-selective tetrasubstituted olefins substituted with a heteroatom of sulfur. This reaction tolerates a broad range of functional groups. In addition, the intermediate trapping experiment and the reaction of optically active allenoate confirmed the presence of sulfonyl and allylic radicals. Cyclic voltammetry studies and Stern−Volmer quenching experiments demonstrated that the reaction proceeds via an oxidative quenching mechanism. $Cu(acac)_2$ was found to play a crucial role in the formation of stereodefined π-allyl Cu species in stereoselective chlorosulfonylation.



INTRODUCTION

Stereodefined tetrasubstituted alkenes are of significant interest due to their unique biological, material, and chemical properties. The distinct biological activities of their *Z* and *E* isomers have been well recognized in medicinal chemistry; for instance, (*E*)-clomiphene exhibits antiestrogenic activity, whereas (*Z*)-clomiphene displays estrogenic effects (Scheme 1a).[1−3] Fully substituted olefins have been extensively developed for applications in functional materials, including photochromic molecular switches and molecular motors.[4,5] Moreover, they serve as versatile molecular scaffolds in organic synthesis, enabling the construction of complex molecular architectures.[6−10] Such tetrasubstituted olefins[11−15] have been traditionally synthesized via the olefination of ketones,[16−18] olefin metathesis,[19] carbometalation of alkynes,[20−27] or transition metal-catalyzed cross-coupling of preformed stereodefined alkenyl halides or their equivalents;[28−30] however, the challenge is the control of the stereo- and regioselectivity. These strategies always provide all carbon-substituted tetrasubstituted olefins. Recently, much attention has been paid to the stereodefined tetrasubstituted alkenes with a C−B bond due to the importance of the boron atom in medicinal chemistry: Song and co-workers developed a copper-catalyzed borylalkynylation of 1,3-diynes with B_2pin_2 and bromoacetylene, enabling the selective synthesis of boronated phenyldiethynylethylene (BPDEE) derivatives.[31] Recently, a practical stereoselective method for synthesizing fully substituted stereodefined alkenyl boron compounds via electrophile-induced 1,2-migration of alkynyl-9-BBN boronates has been reported by Aggarwal et al.[32]

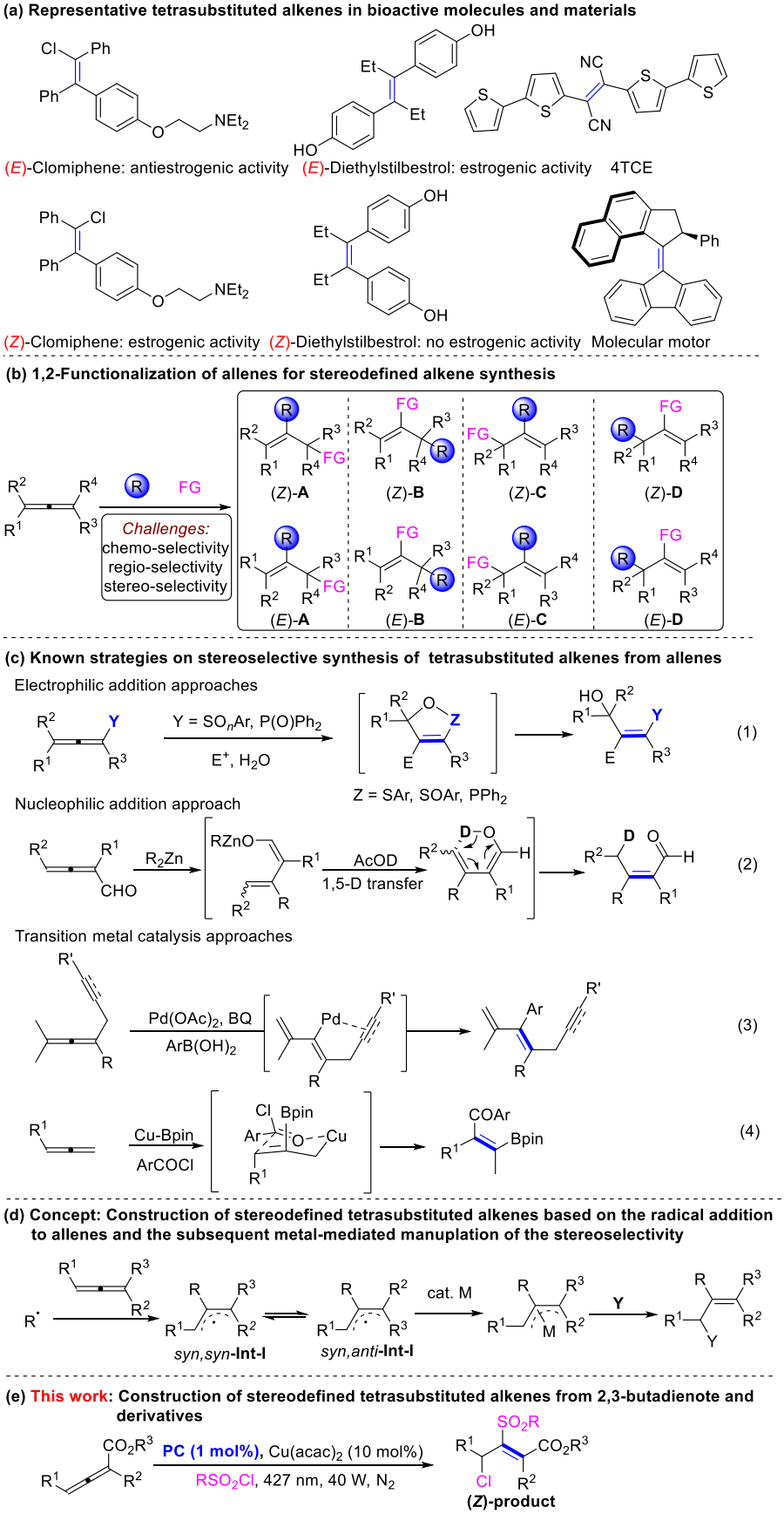
On the other hand, allenes are a class of compounds with two cumulated C═C bonds, and the functionalization of one of the two C═C bonds in an allene unit would offer a potential route to the efficient synthesis of tetrasubstituted olefins,[33−36] provided that the regio- and stereoselectivity illustrated in Scheme 1b may be addressed.[37−39] For example, the electrophilic hydroxylation of allenyl sulfoxides, sulfones, and phosphine oxides demonstrates an excellent regio- and *E*-selectivity based on the electronic effect and the formation of cyclic intermediates due to the presence of these electron-withdrawing groups with neighboring group participation ability (Scheme 1c-1).[40−42] In 2016, we observed that the conjugated addition of 2,3-allenals with organo zincs afforded stereodefined fully substituted 2-enals via the concerted 1,5-H transfer reaction of the conjugated dienol intermediate (Scheme 1c-2).[43] In 2015, Bäckvall and his co-workers elegantly developed a palladium-catalyzed oxidative arylation of allenes that proceeded with high regio- and stereoselectivity to afford the tetrasubstituted C═C bond by applying a carbon−carbon double or triple bond as the coordination handle (Scheme 1c-3).[44] Another innovative approach to fully substituted *Z*-enones has been demonstrated by Chen and his co-workers using the borylcupration of monosubstituted allenes and the subsequent reaction with benzoyl chloride. The stereoselectivity was controlled by the 6-membered chair form transition state (Scheme 1c-4).[45] On the other hand, the

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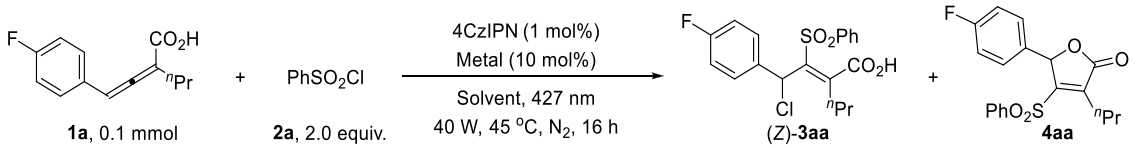
Scheme 1. Useful Representative Tetrasubstituted Alkenes and Tetrasubstituted Alkenes from Allenes



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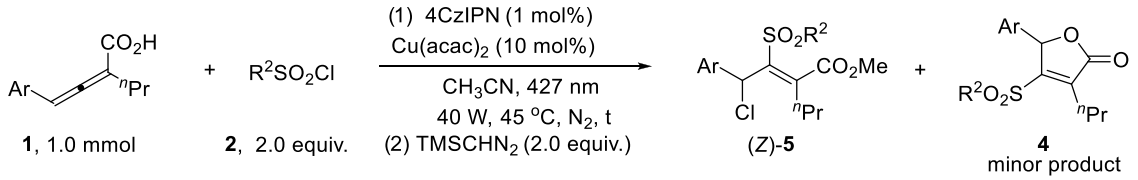
Table 1. Optimization of the Chlorosulfonylation of 2,3-Allenoic Acid 1a with 2a[a]

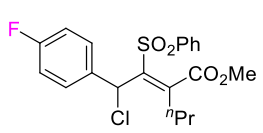


entry	metal	solvent	yield/recovery (%)[b] (Z)-3aa	4aa	(Z)-3aa/4aa	1a
1	-	CH_3CN	4	52	7/93	0
2	$Pd(OAc)_2$	CH_3CN	14	19	42/58	0
3	$Ni(acac)_2$	CH_3CN	47	22	68/32	0
4	$CoCl_2{\cdot}6H_2O$	CH_3CN	53	19	74/26	3
5	$Cu(OAc)_2$	CH_3CN	60	9	87/13	3
6	CuOAc	CH_3CN	25	3	89/11	31
7	$Cu(acac)_2$	CH_3CN	60	8	88/12	14
8	$Cu(acac)_2$	DCM	9	73	11/89	1
9	$Cu(acac)_2$	MeOH	0	0	-	75
10	$Cu(acac)_2$	Ethyl acetate	41	28	59/41	16
11[c]	$Cu(acac)_2$	CH_3CN	52	8	87/13	3
12[d]	**$Cu(acac)_2$**	**CH_3CN**	**69**	**8**	90/10	**4**
13[e]	$Cu(acac)_2$	CH_3CN	66	10	87/13	3

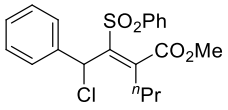
[a]Reaction conditions: allenoic acid **1a** (0.1 mmol), sulfonyl chloride **2a** (0.2 mmol), 4CzIPN (0.001 mmol), and metal catalyst (0.01 mmol) in solvent (2.0 mL) at 45 °C under a N_2 atmosphere. [b]All yields were determined by ^{19}F NMR spectroscopy using trifluorotoluene as an internal standard. [c]1.0 mL of CH_3CN. [d]3.0 mL of CH_3CN. [e]4.0 mL of CH_3CN.

Table 2. Substrate Scope for the Reaction of 2,3-Allenoic Acids 1 with Sulfonyl Chlorides 2[a]

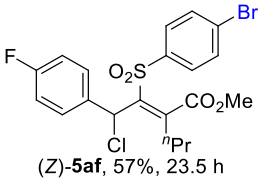




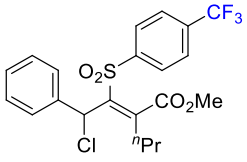
(*Z*)-**5aa**, 59%, 23 h
4aa, 12% NMR yield



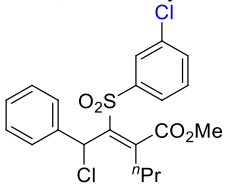
(*Z*)-**5ta**, 58%, 20 h
4ba, 13% NMR yield



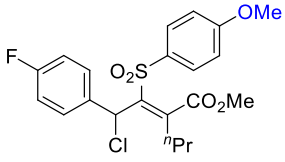
(*Z*)-**5af**, 57%, 23.5 h
4af, 11% NMR yield



(*Z*)-**5th**, 55%, 22 h
4th, 9% NMR yield



(*Z*)-**5td**, 53%, 22 h
4td, 10% NMR yield



(*Z*)-**5aj**, 53%, 23 h
4aj, 12% NMR yield

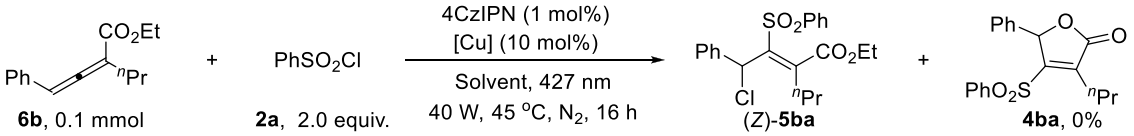
[a]Reaction conditions: 2,3-allenoic acid **1** (1.0 mmol), sulfonyl chloride **2** (2.0 mmol), 4CzIPN (0.01 mmol), and $Cu(acac)_2$ (0.1 mmol) in CH_3CN (30 mL) at 45 °C under a N_2 atmosphere.

radical addition reactions of allenes have been extensively studied;[46−48] however, controlling the stereoselectivity of these reactions is challenging due to the highly reactive nature of the starting and resulting radical species.[49−54] For example, we recently reported a chemo-selective cyclic oxysulfonylation of 2,3-allenoic acids, in which *E*-selective oxysulfonylation products were formed as the major products with *Z*-selective chlorosulfonylation products as the minor products, under the catalysis of an organometallic photocatalyst, $Ir(ppy)_2(dtbbpy)$-PF_6.[55] We envisioned that a metal catalyst may couple with these *in situ* formed nonstereoselective π-allylic radicals to form a stereodefined π-allylic metallic intermediate by applying the template effect of the metal catalyst to tame the reactivity of the radical intermediate, which would be trapped by a reagent of Y to provide a general approach for stereodefined tetrasubstituted olefins (Scheme 1d). Here, as a first attempt at this concept, in this study, the photocatalyzed difunctionalization of 2,3-allenoic acids with sulfonyl chlorides with reversed stereoselectivity was successfully established with the help of a copper catalyst, affording *Z*-selective acyclic

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Table 3. Optimization of Conditions for the Reaction of 2,3-Allenoate 6b with 2a[a]



entry	[Cu]	solvent	yield/recovery (%)[b] (Z)-5ba	yield/recovery (%)[b] 6b
1	$Cu(acac)_2$	CH_3CN	90	0
2	$Cu(acac)_2$	MeOH	0	100
3	$Cu(acac)_2$	DMSO	0	83
4	$Cu(acac)_2$	Toluene	14	0
5	$Cu(OAc)_2$	CH_3CN	64	0
6	$Cu(OTf)_2$	CH_3CN	68	0
7	CuOAc	CH_3CN	80	0
8	$Cu(CH_3CN)_4PF_6$	CH_3CN	76	0
9[c]	**$Cu(acac)_2$**	**CH_3CN**	**90**	**0**
10[d]	$Cu(acac)_2$	CH_3CN	83	0

[a]Reaction conditions: 2,3-allenoate **6b** (0.1 mmol), sulfonyl chloride **2a** (0.2 mmol), 4CzIPN (0.001 mmol), and copper salt (0.01 mmol) in solvent (2.0 mL) at 45 °C under a N_2 atmosphere. [b]All yields were determined by ^1H NMR spectroscopy using CH_2Br_2 as the internal standard. [c]The reaction was conducted at 40 °C. [d]The reaction was conducted at 35 °C.

tetrasubstituted alkenes bearing a sulfur heteroatom as the major products. Furthermore, when 2,3-allenoates were employed instead of 2,3-allenoic acids, this strategy exclusively afforded *Z*-configured tetrasubstituted alkenes under finely tuned conditions (Scheme 1e). It is well-known sulfone unit is commonly observed in biologically active compounds.[56−62] The stereoselective synthesis of such fully substituted alkenyl sulfones will be of immediate biological interest.[63−66]

RESULTS AND DISCUSSION

We began the studies on the reaction of 4-(4-fluorophenyl)-2-propyl-2,3-butadienoic acid **1a** with benzenesulfonyl chloride **2a** in the presence of 4CzIPN in CH_3CN at 45 °C for 16 h. The oxysulfonylation product lactone **4aa** was still observed as the major product in 52% yield, and the *Z*-selective tetrasubstituted acrylic acid (*Z*)-**3aa** was formed in a mere 4% yield (Table 1, entry 1). This is consistent with the fact that the reactions of 2,3-butadienoic acids would readily undergo cyclization to afford lactones.[35,67−71] Interestingly, when a catalytic amount of $Pd(OAc)_2$ was applied as the first attempt of the concept shown in Scheme 1d, the selectivity of (*Z*)-**3aa** over **4aa**[55] reached 42/58 (Table 1, entry 2); surprisingly, upon using $Ni(acac)_2$ and $CoCl_2{\cdot}6H_2O$, such a *Z*/*E* selectivity was reversed (Table 1, entries 3 and 4). With $Cu(acac)_2$ as the cocatalyst, the stereoselectivity was further reversed: the *Z*-selective tetrasubstituted acrylic acid (*Z*)-**3aa** was formed in an optimal yield of 60% as the major product with merely 8% yield of cyclic oxysulfonylation product **4aa** (Table 1, entry 7). Subsequent solvent screening indicated that CH_3CN was still the best (Table 1, entry 7 vs entries 8−10). Conducting the reaction at a concentration of 0.03 M improved the yield of (*Z*)-**3aa** to 69% with a 10% yield of *E*-selective product **4aa** (Table 1, entry 12). Thus, the reaction conditions of entry 12 in Table 1 were defined as the standard conditions for further scope study.

Chlorosulfonylation of 4-(4-fluorophenyl)-2-propyl-2,3-butadienoic acid **1a** with benzenesulfonyl chloride **2a**, followed by esterification with $TMSCHN_2$, afforded the target product (*Z*)-**5aa** in 59% yield with 12% NMR yield of **4aa**. The reaction of 4-phenyl-substituted 2,3-allenoic acid **1t** with **2a** afforded the corresponding product (*Z*)-**5ta** in a decent yield of 58%. In addition, the reactions of differently substituted sulfonyl chlorides with R^2 being bromo- (**2f**), trifluoromethyl- (**2h**), chloro- (**2d**), and methoxy- (**2j**) substituted phenyl groups also provided the desired products (*Z*)-**5af**, (*Z*)-**5th**, (*Z*)-**5td**, and (*Z*)-**5aj** in moderate yields (Table 2).

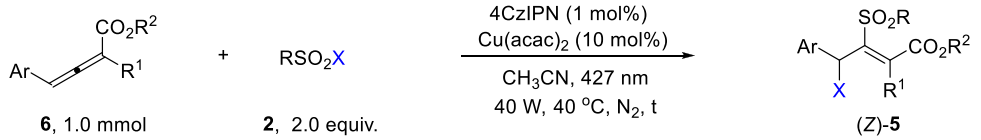
Furthermore, although it is also well established that 2,3-butadienotes would also readily undergo cyclization to afford lactones,[35,67−71] upon using ethyl 4-phenyl-2-propyl-2,3-butadienoate **6b** instead of 4-phenyl-2-propyl-2,3-butadienoic acid **1b**, the (*Z*)-selective product of ethyl 4-chloro-4-phenyl-3-(phenylsulfonyl)-2-propylbut-2(*Z*)-enoate (*Z*)-**5ba** was obtained *exclusively* in 90% yield and the cyclic oxysulfonylation product of 5-phenyl-4-(phenylsulfonyl)-3-propylfuran-2(5*H*)-one **4ba** was not formed under the optimal conditions (Table 3, entry 1). After screening the solvents MeOH, dimethyl sulfoxide (DMSO), and toluene, it was found that none of these performed better than CH_3CN (entries 2−4 vs entry 1). Subsequent copper salt screening indicated that $Cu(acac)_2$ was still the best (Table 3, entries 5−8 vs entry 1). The reaction at 40 °C afforded a similar yield (Table 3, entry 9 vs entry 1). Thus, the reaction conditions of entry 9 in Table 3 were defined as the standard conditions for further scope studies.

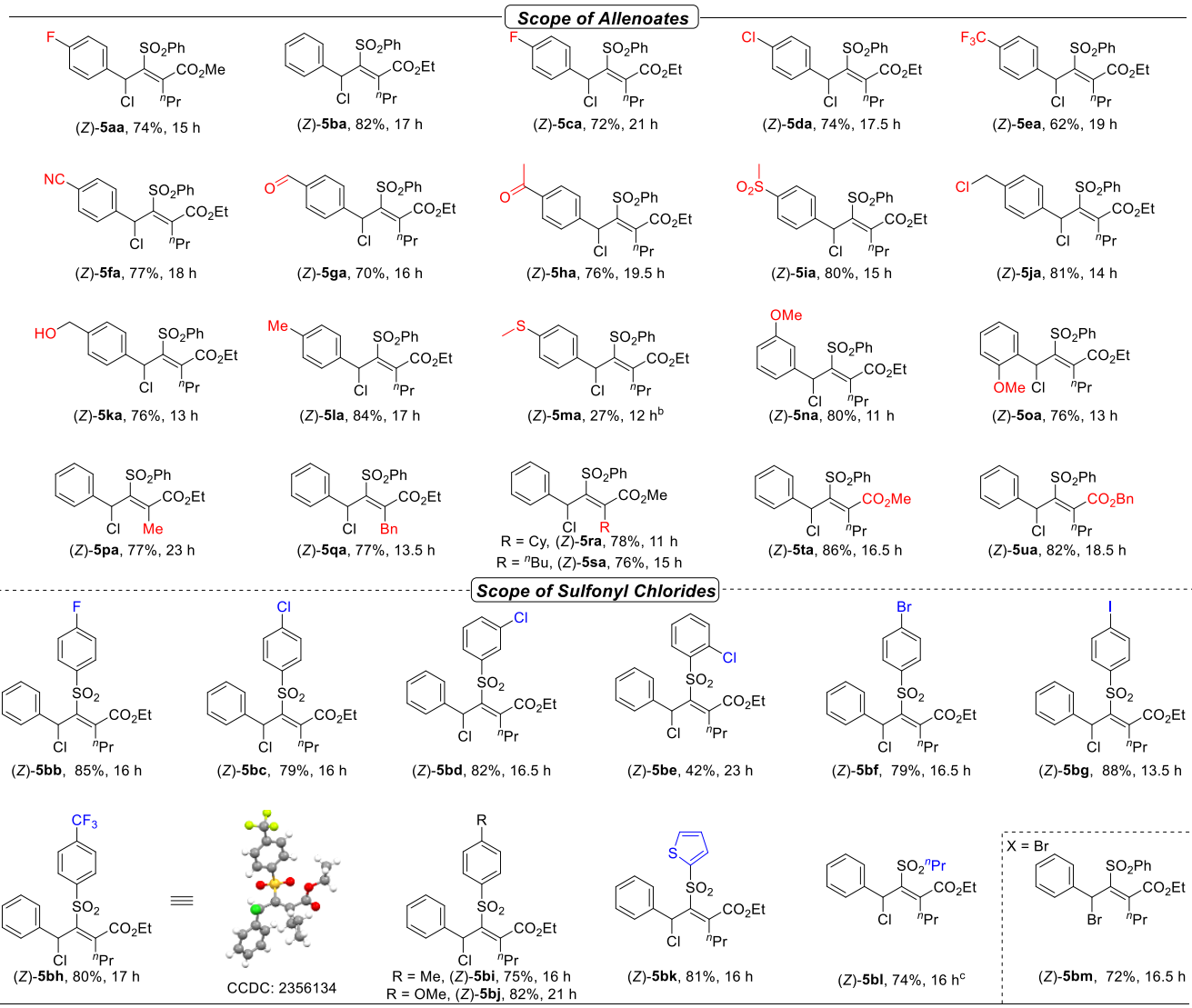
With these optimized conditions in hand, we explored the scope of 2,3-allenoates and sulfonyl chlorides (Table 4). Electron-withdrawing as well as synthetically useful groups such as fluoro, chloro, trifluoromethyl, cyano, aldehyde, ketone, and sulfone may all be incorporated into the 4-aryl group of 2,3-allenoates (**6a**∼**6i**), affording the corresponding products (*Z*)-**5aa**∼(*Z*)-**5ia** in decent yields. Electron-donating group substituted 4-aryl 2,3-allenoates, such as the highly reactive chloromethyl (**6j**), methylthio with a strong coordination ability (**6m**), and methoxy (**6n** and **6o**), were also tolerated, yielding the corresponding products (*Z*)-**5ja** and (*Z*)-**5la**∼(*Z*)-**5oa** in moderate to good yields. To our delight, even the free OH group in **6k** was compatible to deliver the desired product (*Z*)-**5ka** in a 76% yield. The R^1 group may be replaced by methyl (**6p**), benzyl (**6q**), cyclohexyl (**6r**), and *n*-butyl (**6s**). Furthermore, methyl ester (**6t**) and benzyl ester (**6u**) also proceeded smoothly, delivering (*Z*)-**5ta** and (*Z*)-**5ua**

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Table 4. Substrate Scope of the Reaction of 2,3-Allenoates 6 with Sulfonyl Chlorides 2[a]





[a]The reaction was conducted with 2,3-allenoate **6** (1.0 mmol), sulfonyl chloride **2** (2.0 mmol), 4CzIPN (0.01 mmol), and $Cu(acac)_2$ (0.1 mmol) in CH_3CN (20.0 mL) at 40 °C under a N_2 atmosphere. [b]The recovery of **6m** was 7%, as determined by NMR analysis. [c]The recovery of **6b** was 9%, as determined by NMR analysis.

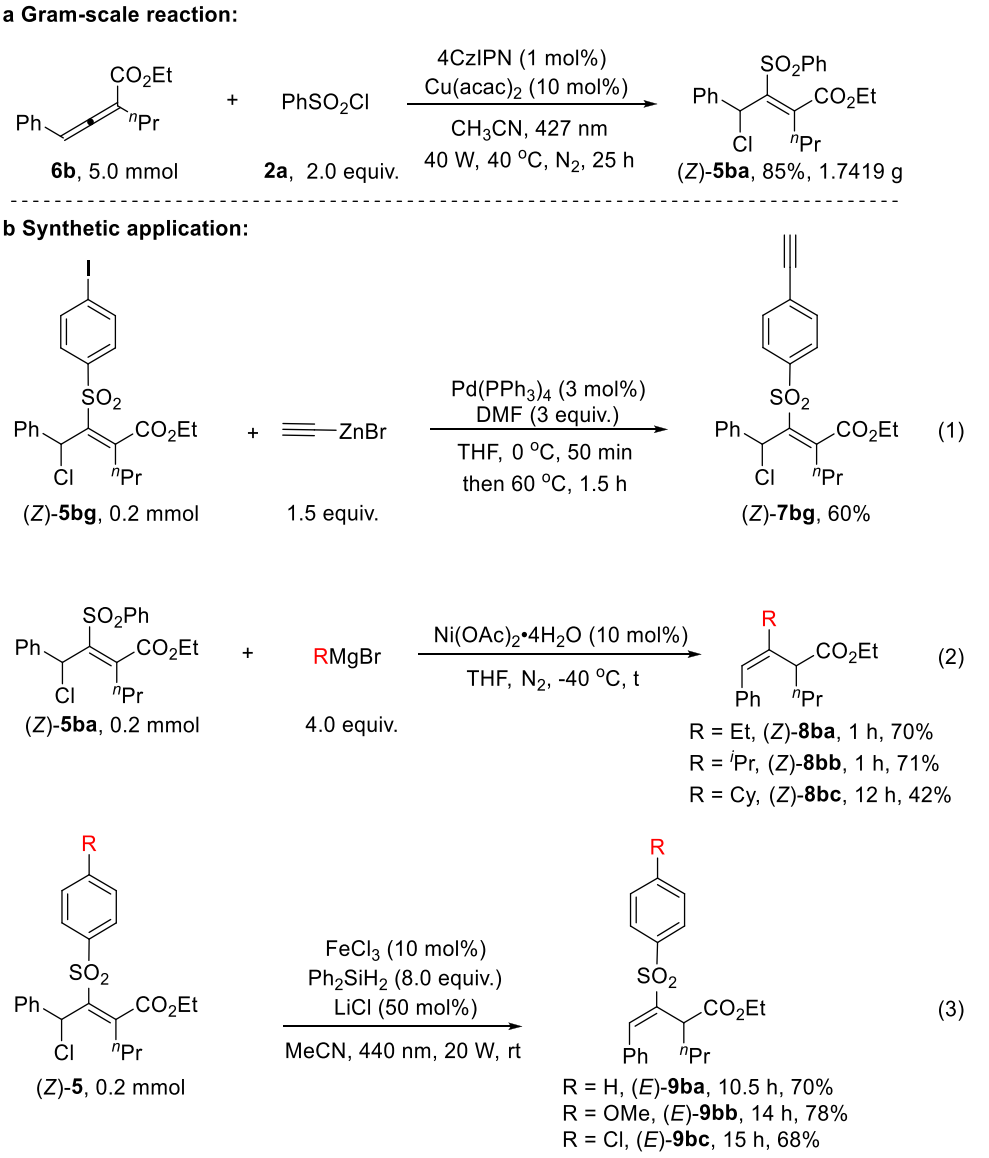
in 86% and 82% yields, respectively. With regard to sulfonyl chlorides, the versatile functional groups such as F, Cl, Br, I, CF_3, or OMe may be incorporated onto the aryl group (the R group), generating the (*Z*)-selective products (*Z*)-**5bb**~(*Z*)-**5bj** in 42%~88% yields. Furthermore, even thienylsulfonyl chloride (**2k**) and *n*-propylsulfonyl chloride (**2l**) also worked, providing the expected products (*Z*)-**5bk** and (*Z*)-**5bl** in 81% and 74% yields, respectively. Benzenesulfonyl bromide **2m** could react with allenoate **6b** to form the target bromosulfonylation product (*Z*)-**5bm** in 72% yield. The structure of the *Z*-selective chlorosulfonylation product was further confirmed by the single-crystal X-ray diffraction study of compound (*Z*)-**5bh**.

A gram-scale reaction was executed under the standard conditions, affording (*Z*)-**5ba** in 85% yield. The carbon−iodine bond in the chlorosulfonylation product (*Z*)-**5bg** underwent the Negishi coupling to give ethyl 4-chloro-3-((4-ethynylphenyl)sulfonyl)-4-phenyl-2-propylbut-2(*Z*)-enoate (*Z*)-**7bg** in 60% yield.[72] Treatment of (*Z*)-**5ba** with ethylmagnesium bromide delivered ethyl 3-ethyl-4-phenyl-2-propylbut-3(*Z*)-enoate (*Z*)-**8ba** in 70% yield in the presence of the $Ni(OAc)_2{\cdot}4H_2O$ catalyst.[73] Similarly, the corresponding product (*Z*)-**8bb** or (*Z*)-**8bc** could be obtained in 71% or 42% yields, respectively, when isopropylmagnesium bromide or cyclohexylmagnesium bromide was employed. Moreover, (*Z*)-**5ba** could be readily converted to ethyl 4-phenyl-3-phenyl-

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Scheme 2. Gram-Scale Reaction and Derivatization



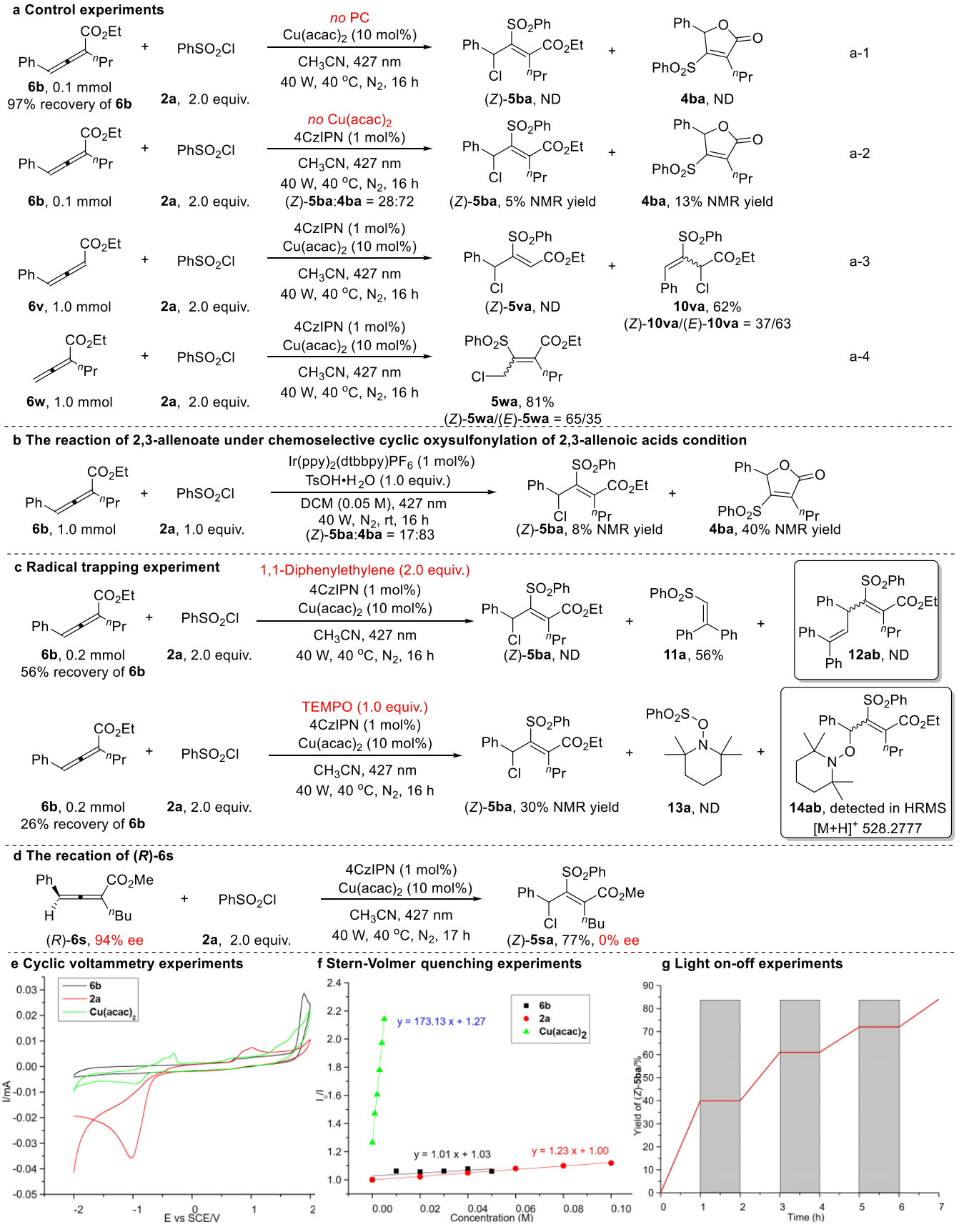
sulfonyl-2-propylbut-3(*E*)-enoate (*E*)-**9ba** in 70% yield with $FeCl_3$ as the catalyst. This transformation could also be applied to other substrates, providing (*E*)-**9bb** or (*E*)-**9bc** in 78% or 68% yield (Scheme 2).[74]

To gain insight into the mechanism, first, the reaction of ethyl 4-phenyl-2-propyl-2,3-butadienoate **6b** with $PhSO_2Cl$ in the absence of 4CzIPN failed to afford (*Z*)-**5ba** with 97% recovery of **6b** (Scheme 3, a-1). In contrast, the reaction of **6b** with $PhSO_2Cl$ in the absence of $Cu(acac)_2$ provided the chlorosulfonylation product (*Z*)-**5ba** in 5% yield with 13% cyclic oxysulfonylation product **4ba** (Scheme 3, a-2). These results indicate that both 4CzIPN and $Cu(acac)_2$ are crucial for the transformation. However, the reaction of ethyl 4-phenyl-2,3-butadienoate **6v** (R^1 is a hydrogen atom) with $PhSO_2Cl$ **2a** afforded an unexpected *Z*/*E* mixture (37/63) of the regioisomeric 2-chloro-substituted chlorosulfonylation product **10va**, and the expected chlorosulfonylation product (*Z*)-**5va** was not observed (Scheme 3, a-3). The reaction of 1,1′-disubstituted allenoate, ethyl 2-propyl-2,3-butadienoate **6w**, with **2a** afforded a *Z*/*E* mixture of the normal chlorosulfonylation product **5wa** in 81% yield (65/35) (Scheme 3, a-4). These results indicated that both the aromatic ring at the 4-position and the alkyl group at the 2-position of the allenoates are crucial to this transformation. When 2,3-allenoate **6b** was subjected to the original photocatalytic cyclic oxysulfonylation conditions,[55] 40% of cyclization product **4ba** was obtained as the major product with 8% of difunctionalization product (*Z*)-**5ba** as the byproduct (Scheme 3b). This result also demonstrates the importance of the 4CzIPN and $Cu(acac)_2$ system in the generation of chlorosulfonyl-containing product (*Z*)-**5**. Second, the reaction of **6b** with **2a** in the presence of 1,1-diphenylethylene as a radical trapping reagent furnished 1,1-diphenyl-2-benzenesulfonylethylene **11a** in 56% yield, and the radical trapping product ethyl 4,6,6-triphenyl-3-(phenylsulfonyl)-2-propylhexa-2,5-dienoate **12ab** was not observed. Furthermore, when the reaction of **6b** with **2a** was conducted in the presence of TEMPO as a radical trapping reagent, the chlorosulfonylation product (*Z*)-**5ba** was formed in only 30% NMR yield with 26% recovery of **6b**, and the TEMPO adduct of the allyl radical **14ab** was observed in HRMS (Scheme 3c). When chiral 2,3-allenoate (*R*)-**6s** was subjected to the standard conditions, the desired product (*Z*)-**5sa** was obtained in 77%

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Scheme 3. Mechanistic Studies

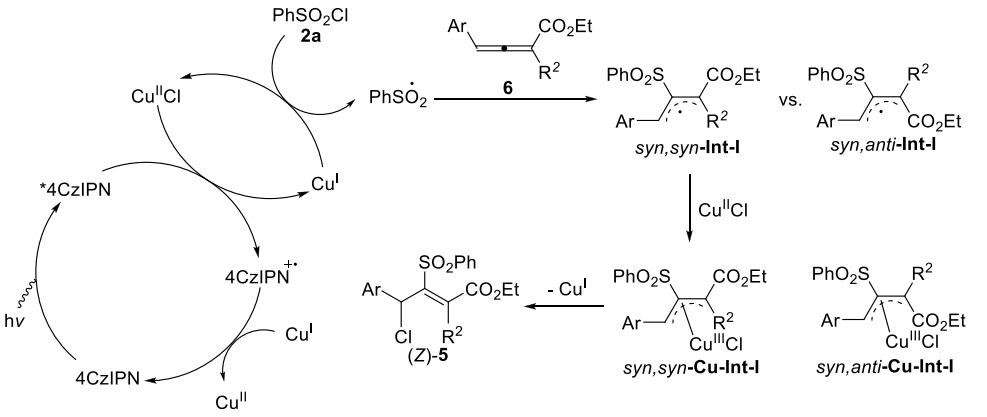


yield, however, in a racemic form (Scheme 3d). These results indicate the presence of sulfonyl and allylic radicals. Finally, cyclic voltammetry experiments confirmed that both $Cu(acac)_2$ ($E_{1/2}^{red}$ $[Cu(acac)_2/Cu(acac)_2^{\bullet-}]$ = −0.77 V vs SCE in

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Scheme 4. Proposed Mechanism



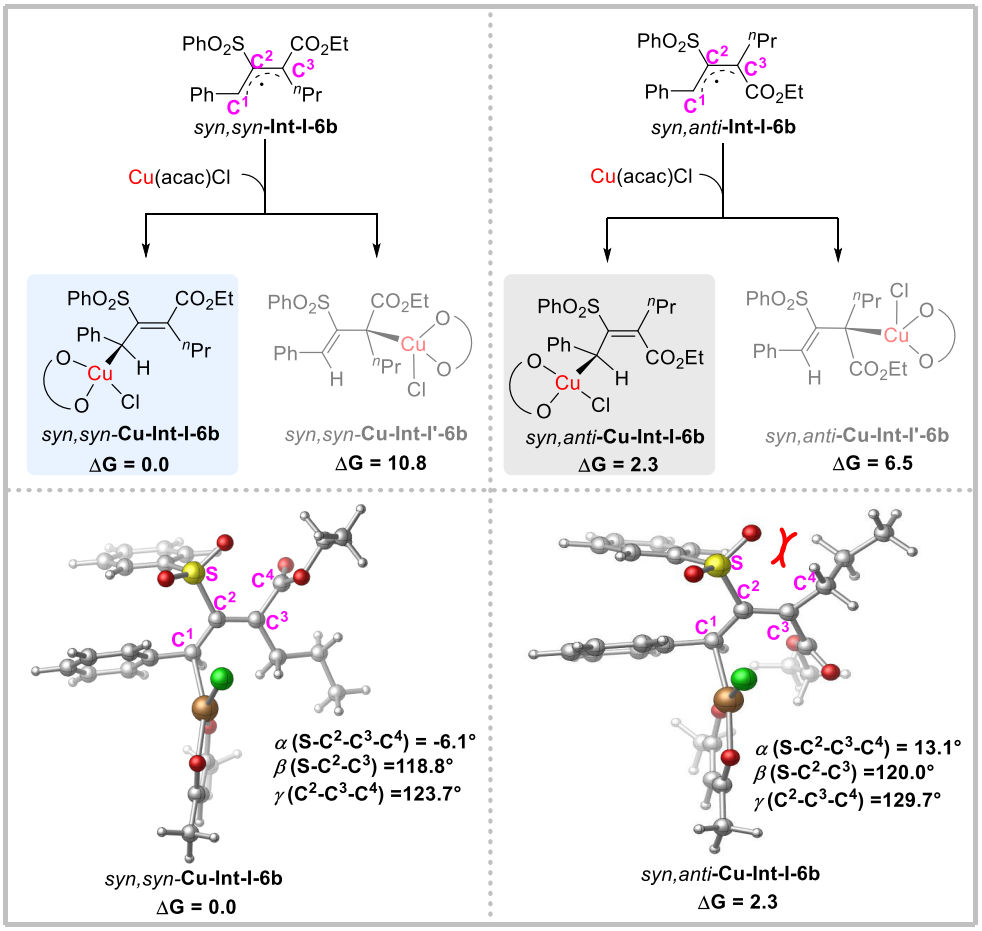


Figure 1. Stability of *syn,syn*-**Cu-Int I** evaluated by DFT calculations at the M06/6-311+G(d,p)-SDD(Cu)-SMD (acetonitrile) // B3LYP-D3(BJ)/6-31G(d,p)-LANL2DZ(Cu) level of theory (Free energies are in kcal/mol)

CH_3CN) and benzenesulfonyl chloride **2a** ($E_{1/2}^{red}$ [**2a**/**2a**$^{\bullet-}$] = −0.84 V vs SCE in CH_3CN) could be reduced by the excited photocatalyst *4CzIPN ($E_{1/2}^{red}$ [4CzIPN$^{\bullet+}$/*4CzIPN] = −1.04 V vs SCE in CH_3CN) (Scheme 3e).[75] It should be noted that no reduction potential peak was observed from −2 to 2 V for ethyl 4-phenyl-2-propyl-2,3-butadienoate **6b**, suggesting that **6b** had a high reduction potential, which was beyond the reduction window of *4CzIPN (Scheme 3e); Stern−Volmer quenching studies showed that $Cu(acac)_2$ could quench the excited photocatalyst *4CzIPN more efficiently than **2a** and **6b** (Scheme 3f), demonstrating $Cu(acac)_2$ would be the exact quencher of the excited photocatalyst; light on−off experiments disclosed that the light was indispensable and the chain reaction mechanism was excluded (Scheme 3g).[76]

Based on these experimental data, a possible mechanism is depicted in Scheme 4. The excited photocatalyst *4CzIPN could be oxidatively quenched with Cu^{II} species to generate the 4CzIPN$^{\bullet+}$ and Cu^{I} species. The Cu^{I} species could react with sulfonyl chloride **2a** to regenerate the Cu^{II} species and sulfonyl radical via a single-electron transfer process. Subsequently, the sulfonyl radical interacted with 2,3-allenoates **6** to afford allyl radical intermediates: *syn,syn*-**Int I**

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and *syn,anti*-**Int I**, which would be trapped by the Cu^{II} species to afford the more stable allyl Cu^{III} species of *syn,syn*-**Cu-Int I**. Then, reductive elimination would deliver the desired chlorosulfonylation product (*Z*)-**5** and regenerate the Cu^{I} species. At the same time, 4CzIPN was regenerated by the interaction of $4CzIPN^{\bullet+}$ with the Cu^{I} species to complete the catalytic cycle (Scheme 4). It should be noted that in the reaction of 2,3-butadienoic acid, the cyclized product **4** could be observed as a minor product because the generated η^3-allylic intermediate is susceptible to intramolecular attack by the carboxylic acid.

The regioselectivity and relative stability of the key allyl–Cu^{III} intermediates were further corroborated by DFT calculations performed on model substrate **6b** (Figure 1). Cu^{II}(acac)Cl selectively traps the radical intermediates at the less sterically hindered C^1 position to form the allyl-Cu^{III} species *syn,syn*-**Cu-Int-I-6b** and *syn,anti*-**Cu-Int-I-6b**. In contrast, the energies of the C^3-attacked counterparts (*syn,syn*-**Cu-Int-I'-6b** and *syn,anti*-**Cu-Int-I'-6b**) are significantly higher by 10.8 and 6.5 kcal/mol, respectively, due to steric repulsion. Furthermore, the energy difference of 2.3 kcal/mol between the two C^1-attacked intermediates (*syn,syn*-**Cu-Int-I-6b** and *syn,anti*-**Cu-Int-I-6b**) arises from the greater structural distortion in *syn,anti*-**Cu-Int-I-6b**, originating from the interaction between the sulfonyl and *n*-propyl groups. This is evidenced by a comparative geometric analysis: in *syn,anti*-**Cu-Int-I-6b**, the substituents of the $C^2{=}C^3$ double bond deviate more from planarity, as reflected in the $S{-}C^2{-}C^3{-}C^4$ dihedral angle (−6.1° in *syn,syn* vs 13.1° in *syn,anti*). Additionally, the $C^2{-}C^3{-}C^4$ bond angle widens from 123.7° in *syn,syn*-**Cu-Int-I-6b** to 129.7° in *syn,anti*-**Cu-Int-I-6b**. These structural distinctions clearly account for the observed energy gap between the two intermediates.

CONCLUSION

In summary, a photo/copper dual catalysis has been established for the chemo-, regio-, and stereoselective chlorosulfonylation reaction of 2,3-allenoates with sulfonyl chlorides to afford *Z*-specific tetrasubstituted alkenes with sulfonyl group substitution. This protocol accommodates synthetically useful functional groups such as halogens, trifluoromethyl, cyano, aldehyde, ketone, sulfone, hydroxyl, methylthio, and methoxy. It is noted that $Cu(acac)_2$ is crucial for stereoselective chlorosulfonylation via the formation of stereodefined allylic copper intermediates. Cyclic voltammetric and Stern–Volmer quenching experiments demonstrated that the excited photocatalyst *4CzIPN was oxidatively quenched by Cu^{II} species. Radical trapping experiments and data on optically active 2,3-allenoate revealed that sulfonyl and allylic radicals were generated in this catalytic cycle. Further studies are being pursued in our laboratory.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/jacs.5c17536.

Detailed experimental procedures; characterization of new compounds (PDF)

Accession Codes

Deposition Number 2356134 contains the supporting crystallographic data for this paper. These data can be obtained free of charge via the joint Cambridge Crystallographic Data Centre (CCDC) and Fachinformationszentrum Karlsruhe Access Structures service.

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Notes

The authors declare no competing financial interest.

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